

REQUIREMENTS ANALYSIS & SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Commercial Livestock Marketplace and Record-Keeping Application

1. Project Introduction

The proposed project is a commercial marketplace and record-keeping application intended to improve access to livestock, crop and animal feed market for farmers, buyers and sellers. The application will provide a platform where livestock can be advertised and viewed by potential buyers, while also allowing farmers to keep important records about their animals.

During the stakeholder meeting, the main focus identified for the project was access to the market. The stakeholder explained that farmers should be able to reach a wider range of buyers and that buyers should be able to find livestock from different farms through one platform. The application should support commercial livestock activities and provide opportunities for users to connect with markets at local, national and potentially international levels.

The application will also assist farmers with keeping basic information about their livestock. This includes information such as an animal's date of birth, parents and vaccination history. Sellers will be able to provide information such as the animal's type, weight, price and brandmark. The application will also allow sellers to upload documentation as proof that they own the livestock being offered for sale.

The main users of the application will be buyers and sellers. Sellers will use the platform to advertise their livestock, while buyers will use it to browse available livestock and find animals that meet their requirements.

2. Stakeholder Profile and Current Process

2.1 Organization and Stakeholder Context

The stakeholder operates within the commercial livestock sector and identified the need for a application that can improve the way farmers access and participate in the livestock market.

The main business need is to create a platform that connects livestock sellers with potential buyers and makes it easier for users to find livestock beyond their immediate farming networks. The application should also provide relevant information about the livestock so that buyers can make informed decisions.

2.2 Stakeholder

The primary stakeholder is the representative who provided the requirements for the proposed livestock marketplace. During the first stakeholder engagement, the stakeholder explained the current needs of the livestock market and identified the main functions that the proposed application should provide.

The stakeholder emphasized that the project is commercially focused and that access to the market should remain the main purpose of the application.

2.3 Users

Sellers/Farmers

- Record information about their livestock.
- List livestock that is available for sale.
- Enter the price of livestock.
- Provide the type or breed of livestock.
- Provide the weight of the livestock.
- Provide landmark information.
- Upload documentation showing ownership of livestock.
- Make their livestock visible to potential buyers.

Buyers

- Browse livestock that is available for sale.
- View livestock from different farms/locations .
- View information about individual livestock.
- Check the weight and price of livestock.
- Find livestock that matches their requirements.
- Access livestock listings from different sellers.

2.4 Existing Process

From the stakeholder discussion, it was identified that farmers need a better way of accessing the livestock market and reaching potential buyers. Buyers also need a convenient way of finding livestock that is available from different farms.

The proposed application will provide a central digital platform where this information can be brought together. Instead of relying only on existing contacts or individual farm networks, sellers will be able to make their livestock available to a wider group of potential buyers.

2.5 Identified Problems

1. Farmers need improved access to the livestock market.
2. Sellers need a way to reach more potential buyers.
3. Buyers need an easier way to find livestock from different farms.
4. Important information about individual livestock needs to be recorded and maintained.
5. Buyers need access to information such as livestock weight, type and price.
6. Sellers need a way to provide evidence of livestock ownership.
7. Landmark information needs to be associated with the livestock being offered.

8. Farmers need opportunities to connect with wider national and potentially international markets.

3. Problem Statement

3.1 Current Situation

The livestock sector depends on effective connections between farmers who want to sell livestock and buyers who are looking for specific animals. Farmers may have difficulty reaching buyers outside their existing networks, while buyers may not have an easy way of finding livestock available from different farms.

There is also a need to keep relevant information about individual animals, including their date of birth, parents, vaccination history, weight and ownership information.

3.2 Problem

The main problem identified by the stakeholder is the lack of a centralised digital platform that provides easier access to the commercial livestock market while making relevant livestock information available to buyers and sellers.

3.3 Affected Users

- Livestock farmers and sellers.
- Livestock buyers.
- Commercial livestock businesses.
- Farmers looking for livestock from other farms.
- Potential buyers and sellers operating across wider markets.

3.4 Consequences

If farmers do not have access to a suitable platform, they may have difficulty reaching a wider range of buyers. Buyers may also spend more time trying to find livestock that meets their requirements.

Limited access to information can make the process more difficult because buyers may not have immediate access to details such as the animal's weight, type, price or ownership information. This can also limit opportunities for farmers who want to expand their market beyond their existing contacts.

3.5 Need for Improvement

There is a need for a commercial digital platform that brings livestock buyers and sellers together and provides relevant information about livestock in one place. The platform should make it easier for sellers to advertise their livestock and for buyers to find livestock from different farms.

4. Project Aim and Objectives

4.1 Project Aim

The aim of the project is to develop a commercial livestock marketplace and record-keeping application that improves access to the livestock market by connecting buyers and sellers and providing relevant information about livestock.

4.2 Project Objectives

1. Develop a digital marketplace for commercial livestock.
2. Allow sellers to advertise livestock for sale.
3. Allow buyers to view livestock available from different farms.
4. Provide information such as livestock type, weight and price.
5. Allow farmers to record basic information about individual animals.
6. Allow farmers to record an animal's date of birth, parents and vaccination information.
7. Allow sellers to provide landmark information.
8. Allow sellers to upload documentation showing ownership of livestock.
9. Improve the ability of farmers to reach a wider livestock market.
10. Support connections between buyers and sellers at national and potentially international levels.
11. Provide a platform that can support commercial livestock activities.

5. Project Scope 5.1

In-Scope

User Management

- Registration for buyers and sellers.
- User access based on the user's role.
- Basic user profile information.

Livestock Marketplace

- Creating livestock listings.
- Displaying livestock listings.
- Providing the price of livestock.
- Providing livestock type or breed information.
- Displaying livestock weight.
- Allowing buyers to view livestock from different farms.
- Allowing sellers to make livestock available to potential buyers.

Livestock Records

The application will allow farmers to record basic information about individual animals, including:

- Date of birth.
- Parents.
- Type or breed.
- Vaccination information.
- Weight.

Ownership and Brandmark

- Provide brandmark information.
- Upload documentation showing ownership of livestock.

Commercial Market Access

The application will focus on connecting buyers and sellers and improving access to the livestock market. It will also provide a foundation for access to national and potentially international markets.

5.2 Deliverables

1. A functional livestock marketplace application.
2. Buyer functionality.
3. Seller/farmer functionality.
4. Livestock listing functionality.
5. Livestock record-keeping functionality.
6. Ownership documentation upload functionality.
7. Brandmark information functionality.
8. A database for storing relevant user and livestock information.
9. An integrated application using Flutter, ASP.NET and Supabase.

5.3 Acceptance Criteria

- A user can register and access the application.
- Sellers can create livestock listings.
- Sellers can provide the price of livestock.
- Sellers can provide livestock type and weight.
- Farmers can record basic information about an animal.
- Birth, parent and vaccination information can be stored.
- Sellers can provide brandmark information.
- Sellers can upload ownership documentation.
- Buyers can view available livestock.
- Buyers can view livestock from different farms.
- Relevant livestock information can be stored and retrieved.

5.4 Exclusions

The following functions have not yet been fully defined by the stakeholder and will therefore not be treated as confirmed requirements for the initial implementation:

- The exact auction or bidding process.
- Online payment processing.
- Livestock transportation and delivery management.
- International payment and currency conversion.
- Formal verification of ownership documents.
- Formal verification of brandmarks.
- Insurance services.
- Additional livestock health services beyond the information specifically mentioned by the stakeholder.

These functions may be considered in a later phase if the stakeholder confirms that they are required.

5.5 Constraints

- The time available for development.
- The resources available to the development team.
- The technical skills and experience of the team.
- Changes to requirements following further stakeholder meetings.
- The need to protect user and livestock information.
- The technical complexity involved in supporting national and international users.
- The integration of Flutter, ASP.NET and Supabase into one solution.

5.6 Assumptions

1. Users will have access to the internet when using the application.
2. Sellers will be responsible for providing accurate information about their livestock.
3. Sellers will provide legitimate ownership documentation.
4. Users will provide accurate information when registering.
5. The first version will focus on the core marketplace and livestock record-keeping functions.
6. Requirements that were not fully explained during the first meeting will be clarified during future stakeholder engagement.
7. Payment, delivery and detailed auction functionality will only be included if the stakeholder confirms these requirements and they are feasible within the project.

6. Stakeholder Needs

The stakeholder engagement identified the following needs:

Stakeholder Need	Priority
Improve access to the livestock market.	High
Allow farmers to advertise livestock commercially.	High
Allow buyers to find livestock from different farms.	High
Provide information about livestock weight.	High
Provide livestock type or breed information.	High
Provide livestock price information.	High
Allow farmers to record an animal's date of birth.	Medium
Allow farmers to record an animal's parents.	Medium
Allow farmers to record vaccination information.	Medium
Allow sellers to provide landmark information.	High
Allow sellers to upload documentation showing ownership of livestock.	High
Connect buyers and sellers across a wider market.	High
Support feedlot cattle weighing 220 kg and above.	High
Provide the ability to sell livestock-related products, subject to further clarification.	Medium

The stakeholder made it clear that access to the market is the main focus of the proposed application.

7. Functional Requirements

Functional requirements describe the functions that the proposed application must provide.

7.1 User Registration and Access

FR-01: The app will allow users to register for an account.

FR-02: The application will allow users to use the app as either a buyer or seller.

FR-03: The app will provide secure access to registered users.

7.2 Seller Functionality

FR-04: The app will allow sellers to create livestock listings.

FR-05: The app will allow sellers to enter the price of livestock.

FR-06: The app will allow sellers to enter the type or breed of livestock.

FR-07: The app will allow sellers to enter the weight of livestock.

FR-08: The app will allow sellers to provide landmark information.

FR-09: The app will allow sellers to upload documentation showing ownership of livestock.

FR-10: The app will allow sellers to manage the livestock information associated with their listings.

7.3 Livestock Record-Keeping

FR-11: The app will allow farmers to record the date of birth of an animal.

FR-12: The app will allow farmers to record the parents of an animal.

FR-13: The app will allow farmers to record vaccination information.

FR-14: The app will store and retrieve relevant livestock information.

7.4 Buyer Functionality

FR-15: The app will allow buyers to view available livestock listings.

FR-16: The app will allow buyers to view livestock from different farms.

FR-17: The app will display relevant information about listed livestock.

FR-18: The app will display the weight of listed livestock.

FR-19: The app will display the price of listed livestock.

FR-20: The app will provide buyers with access to livestock offered by different sellers.

7.5 Market Access

FR-21: The app will provide a platform where buyers and sellers can connect.

FR-22: The app will support livestock listings from different farms.

FR-23: The app will support the commercial marketing of livestock.

FR-24: The app will support feedlot livestock requirements, including cattle weighing 220 kg and above.

FR-25: The app should support access to the national livestock market.

FR-26: The app should provide a foundation for future access to international livestock markets.

7.6 Livestock-Related Products

FR-27: The app should support the listing and sale of livestock-related products, subject to further clarification from the stakeholder.

7.7 Auction Functionality

During the stakeholder meeting, the stakeholder compared the desired marketplace experience to an online marketplace and referred to an auction concept where livestock can be placed with a specific price.

The exact process for the auction functionality was not defined during the first meeting. Therefore, the requirement will be kept open for further clarification.

FR-28: The app should allow livestock to be placed on the marketplace with a specified price.

FR-29: The exact auction or bidding process shall be confirmed with the stakeholder during the next requirements consultation.

8. Non-Functional Requirements

8.1 Usability

NFR-01: The app shall provide a simple and user-friendly interface that can be used by both buyers and farmers.

NFR-02: Important livestock information, including price, weight and type, shall be clearly presented to users.

NFR-03: The app should minimize unnecessary steps when creating or viewing livestock listings.

8.2 Performance

NFR-04: The app should respond to normal user actions within a reasonable time under normal operating conditions.

NFR-05: Livestock listings and relevant information should load efficiently for users with a suitable internet connection.

8.3 Security

NFR-06: User accounts shall be protected through secure authentication.

NFR-07: Users shall only have access to functions appropriate to their role.

NFR-08: Uploaded ownership documents shall be protected from unauthorised access.

NFR-09: User and livestock information shall be stored securely.

8.4 Availability

NFR-10: The app should be available to users when the application and its supporting services are operational.

NFR-11: The app should minimize the risk of losing information when users submit livestock records.

8.5 Reliability

NFR-12: The app should reliably store livestock information after successful submission.

NFR-13: The app should retrieve stored livestock information accurately.

8.6 Scalability

NFR-14: The app should be designed to accommodate additional users, farms and livestock listings as the marketplace grows.

NFR-15: The app should provide a foundation for expanding from local users to national and potentially international users.

9. Integration Requirements

9.1 Technology Stack

The application will use the following technologies as specified by the development team:

- Frontend: React Native mobile app UI
- Backend: Node.js business logic & APIs
- Database: Supabase (PostgreSQL) data storage & authentication

The proposed solution will use Flutter, ASP.NET and Supabase. Each technology will perform a different role within the overall application.

9.2 Flutter

Flutter will be used to develop the application's user interface and client-side functionality.

- Displaying the application screens.
- Providing forms for entering livestock information.
- Displaying livestock listings.
- Allowing buyers and sellers to interact with the application.
- Sending requests to the backend.

9.3 ASP.NET

ASP.NET will be used for the backend/API of the application. It will provide the connection between the application's interface and the required backend services.

- Processing requests from the application.
- Handling the app's business logic.
- Managing communication between the application and backend services.
- Processing relevant livestock and marketplace operations.
- Controlling access to backend functionality.

9.4 Supabase

Supabase will provide backend services such as authentication, database services and storage.

- User authentication.
- Storage of livestock records.
- Storage of marketplace listing information.
- Storage of relevant user information.

- Storage of uploaded documentation where required.

9.5 Integration Flow

The proposed integration can be represented as:

Flutter Application → ASP.NET Backend/API → Supabase Services and Database

The user will interact with the Flutter application. When the user performs an action, such as creating a livestock listing, the application will communicate with the ASP.NET backend. The backend will process the request and communicate with Supabase where necessary. The result will then be returned to the Flutter application and presented to the user.

This approach separates the user interface, application logic and data services while allowing the three technologies to operate together as one system.

10. Requirements Change Log

The first stakeholder meeting was used to identify the initial requirements for the proposed application. At this stage, no major stakeholder requirements have been removed.

However, several requirements were mentioned without enough detail to define their complete functionality. These items will need to be discussed with the stakeholder during the next consultation.

Requirement	Reason for Further Clarification	Status
Auction/bidding process	The stakeholder mentioned an auction concept but did not explain how bidding should work.	To be confirmed
Online payments	No payment method or payment process was specified.	To be confirmed
Livestock delivery/transport	The stakeholder did not specify whether transportation should be managed by the application.	To be confirmed
International transactions	International market access was mentioned, but the exact requirements were not discussed.	To be confirmed
Ownership document verification	Document upload was requested, but the process for checking documents was not explained.	To be confirmed
Brandmark verification	Brandmark information was requested, but the verification process was not defined.	To be confirmed
Livestock-related products	The stakeholder mentioned selling livestock-related items, but the exact products or categories were not identified.	To be confirmed

These areas will be reviewed after the next stakeholder consultation and may be changed based on the stakeholder's feedback, feasibility analysis and project constraints.

11. Refined Project Scope

Based on the first stakeholder consultation, the initial scope of the project will focus on developing a commercial livestock marketplace with basic livestock record-keeping functionality.

The core application will allow buyers and sellers to access the platform. Sellers will be able to create livestock listings and provide information such as the livestock type, weight, price and landmark. Sellers will also be able to upload documentation showing ownership of the livestock.

Buyers will be able to view available livestock and access listings from different farms. This will give buyers a central place to find livestock that may previously have been difficult to access through individual farm networks.

The application will also include basic record-keeping functionality. Farmers will be able to record information about individual animals, including their date of birth, parents and vaccination information.

The main focus of the project will remain access to the livestock market. The application will therefore prioritize the connection between buyers and sellers and the ability to make livestock available to a wider market.

Feedlot livestock is also included within the initial scope, with the stakeholder specifically identifying cattle weighing 220 kg and above as relevant.

The application will consider national market access as part of the project direction. International market access will be treated as a future capability until the stakeholder provides more detailed requirements concerning how international users and transactions should be supported.

Advanced functions such as online payments, transportation management, formal document verification and a fully defined auction or bidding system will not be treated as confirmed core functions at this stage. These areas will be discussed with the stakeholder during the next consultation.

The refined scope for the initial application therefore includes:

1. Buyer and seller access.
2. Livestock marketplace functionality.
3. Livestock listing and management.
4. Livestock price information.
5. Livestock type or breed information.
6. Livestock weight information.
7. Landmark information.

8. Ownership documentation upload.
9. Basic livestock record keeping.
10. Animal birth information.
11. Parent information.
12. Vaccination information.
13. Access to livestock from different farms.
14. Commercial livestock market access.
15. Support for feedlot cattle weighing 220 kg and above.
16. A foundation for future expansion into national and international markets.

This scope will be reviewed and updated after further stakeholder engagement.